The Quenched Residue–Height Spectral Shadow

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Remark (Status of these notes). These are working research notes. The construction below separates objects that are already finite and checkable from the one statement that remains a new, unproven theorem; that separation is the whole point of the note, and it is preserved as written. The central *Quenched Spectral-Shadow Principle* is conjectural (the text treats it as something that "would" be proved). Project-internal terms (notably H23a, the quasi-stationary distribution (QSD) machinery, the residue spaces Ω_Q, and the exact affine correction $\varepsilon_{\text{exact}}$) are used without restatement because they refer to a larger body of work. Nothing here has been verified.

1 Keeping the pieces that survive

The useful pieces of the preceding answers are not the proposed names themselves: "Gibbs measure," "arithmetic coboundary sheaf," "spectral-coboundary module," or "Collatz Selberg zeta function." Those are fingers pointing toward the missing object, not the object itself. The pieces worth keeping are the following.

2 The annealed thermodynamic formalism

First, the annealed theory is genuinely unified by the transfer-operator / thermodynamic formalism of the Collatz height cocycle. The basic height increment is

$$\Delta_b = \log_2 3 - b, \qquad b = v_2(3x+1),$$

and under the Haar/annealed valuation law

$$p_b = 2^{-b}, \qquad b \geq 1,$$

the moment transform is

$$M(s) = \sum_{b\geq 1} 2^{-b}\, 2^{s(\log_2 3 - b)} = \frac{3^s}{2^{1+s}-1}.$$

The Cramér root is

$$M(1) = 1,$$

and the critical tilted law is

$$p_b^{(1)} = 2^{-b}\, 2^{\log_2 3 - b} = 3 \cdot 4^{-b} = \tfrac{3}{4}\left(\tfrac{1}{4}\right)^{b-1}.$$

This package explains the annealed height tail, the Cramér tilt, the conditioned high-ascent law, and the pressure side of the killed-walk/QSD machinery.

3 Finite residue-height transfer operators

Second, the finite residue-height transfer operators are the safe spectral objects. For a finite modulus Q and strip width L, one can define a finite state space

$$\mathcal{X}_{Q,L} = \Omega_Q \times I_L,$$

where Ω_Q is the appropriate residue space and I_L is the discretized height/deficit strip. Edges are exact admissible inverse-Syracuse transitions, indexed by valuations b, with residue update

$$a' \equiv 3^{-1}\big(2^b a - 1\big) \pmod{Q},$$

and height update

$$z' = z + \log_2 3 - b + \varepsilon_{\text{exact}}(a, z, b),$$

where $\varepsilon_{\text{exact}}$ records the exact affine correction when it is retained. Killing occurs when z' exits the strip.

For each s, the finite tilted transfer operator is

$$(\mathcal{L}_{s,Q,L} f)(x) = \sum_{e\,:\,y\to x} w_s(e)\, f(y),$$

with edge weight of the form

$$w_s(e) = m(e)\, 2^{s(\log_2 3 - b(e))}.$$

Here $m(e)$ is the exact finite-fiber weight; in the pure annealed model it is $2^{-b(e)}$. The finite determinant

$$\Xi_{Q,L}(z, s) = \det\big(1 - z\mathcal{L}_{s,Q,L}\big)$$

is unambiguously defined. Its logarithm records closed paths in the finite residue-height graph:

$$\log \Xi_{Q,L}(z, s)^{-1} = \sum_{n\geq 1} \frac{z^n}{n}\, \operatorname{tr}\big(\mathcal{L}^n_{s,Q,L}\big).$$

This is the safe finite analogue of a Ruelle or Selberg zeta function. A global S-arithmetic Selberg zeta function is not presently known and should not be assumed.

4 Character sectors and the coboundary obstruction

Third, finite character sectors are the natural home of the coboundary obstruction. Decompose the finite operator by residue characters:

$$\mathcal{L}_{s,Q,L} = \mathcal{L}_{s,Q,L,0} \oplus \bigoplus_{\chi\neq 1} \mathcal{L}_{s,Q,L,\chi}.$$

A nontrivial character χ is a rank-one local system on the finite residue graph. H23a-type statements assert a uniform nontrivial-character gap:

$$\rho\big(\mathcal{L}_{s,Q,L,\chi}\big) \leq (1 - \kappa_Q)\, \rho\big(\mathcal{L}_{s,Q,L,0}\big), \qquad \chi \neq 1.$$

Equivalently, the finite residue graph has no nontrivial local system whose holonomy is invisible to the dynamics. This is the rigorous finite shadow of the phrase "spectral-coboundary obstruction."

5 The quenched residue-height spectral shadow

The object suggested by these pieces is therefore not a single Gibbs measure, not a completed sheaf, and not an already-existing zeta function. It is the following pro-object:

$$\boxed{\mathfrak{Q}_{\mathrm{Coll}} = \Big\{ \mathcal{X}_{Q,L},\ \mathcal{L}_{s,Q,L},\ \mathcal{L}_{s,Q,L,\chi},\ \Xi_{Q,L,\chi},\ \mathscr{C}_{Q,L} \Big\}_{Q,L}}$$

Here $\mathscr{C}_{Q,L}$ denotes the finite local-system/coboundary package generated by the nontrivial residue characters and their holonomies on the exact residue-height graph. More explicitly,

$$\mathscr{C}_{Q,L} = \bigoplus_{\chi \neq 1} \mathbb{C}_\chi,$$

viewed as a coefficient system over the finite directed graph $\mathcal{X}_{Q,L}$. A closed path γ has holonomy

$$\mathrm{hol}_\chi(\gamma) = \prod_{e \in \gamma} \chi(e),$$

and the finite certificate says that for every nontrivial χ, some certified cycle satisfies

$$\mathrm{hol}_\chi(\gamma) \neq 1.$$

Thus nontrivial finite characters cannot behave as coboundaries on the primitive bulk component.

We call the pro-object $\mathfrak{Q}_{\mathrm{Coll}}$ the *quenched residue-height spectral shadow*. It is a shadow because it is not yet the missing theorem itself. It is the finite system in which any future quenched obstruction must become visible.

6 The Quenched Spectral-Shadow Principle

The desired inverse principle can now be stated cleanly.

Quenched Spectral-Shadow Principle (conjectural)

If a deterministic positive-integer orbit violates the annealed height/valuation law persistently, for example by producing an infinite sub-equilibrium tail, repeated critical high ascents, or sustained deviation of

$$D_m - m \log_2 3$$

from the equilibrium line, then there exists a finite pair (Q, L) and a nontrivial character sector $\chi \neq 1$ such that the bad orbit has a nonzero finite spectral shadow in $\mathscr{C}_{Q,L}$. Equivalently, the bad orbit forces an approximate peripheral obstruction:

$$\rho_{\mathrm{eff}}\big(\mathcal{L}_{s,Q,L,\chi}\big) \approx \rho\big(\mathcal{L}_{s,Q,L,0}\big).$$

If this principle were proved, H23a would rule out the obstruction:

$$\rho\big(\mathcal{L}_{s,Q,L,\chi}\big) \leq (1 - \kappa_Q)\, \rho\big(\mathcal{L}_{s,Q,L,0}\big), \qquad \chi \neq 1.$$

Thus the contradiction would be:

$$\text{bad deterministic orbit} \implies \text{finite spectral/coboundary shadow} \implies \text{violation of H23a.}$$

This is the precise form of the missing annealed-to-quenched bridge.

7 Finite versus new

The point of the construction is that it separates what is already finite and checkable from what remains genuinely new.

Component	Status
$\mathcal{L}_s,\ M(s),\ p_b^{(s)}$	annealed thermodynamic object
$\mathcal{L}_{s,Q,L}$	finite exact residue-height operator
$\Xi_{Q,L}(z,s)$	finite determinant / zeta shadow
$\mathscr{C}_{Q,L}$	finite character / coboundary local-system package
H23a gap	finite spectral obstruction exclusion
Quenched Spectral-Shadow Principle	new missing theorem

8 Discarded overclaims

The discarded overclaims are equally important. A Gibbs or SRB measure does not solve the problem, because ergodicity gives almost-everywhere statements, not control of every arithmetically prescribed integer orbit. A global S-arithmetic Selberg zeta function is not presently constructed, and no known trace formula applies to the noninvertible Collatz monoid. A sheaf or coboundary module is only meaningful after a finite residue-height graph, local systems, and restriction maps are specified. Baker theory may eliminate certain cycle-like Diophantine resonances, but it does not by itself prove deterministic equidistribution. Exceptional Lie theory currently supplies no load-bearing object.

9 The object we can construct now

Thus the answer to the question "what is the object?" is:

> The object we can construct now is the pro-system of finite residue-height tilted transfer operators, their character-sector decompositions, their finite Fredholm determinants, and their local-system/coboundary obstruction package:
>
> $$\mathfrak{O}_{\mathrm{Coll}} = \left\{ (\mathcal{X}_{Q,L},\ \mathcal{L}_{s,Q,L},\ \mathscr{C}_{Q,L}) \right\}_{Q,L}.$$
>
> It is the finite spectral shadow in which any bad deterministic orbit must be detected if the annealed-to-quenched inverse theorem is true.

This is the finger, not the moon. The moon would be the theorem

$$\text{bad quenched orbit} \implies \text{nontrivial finite spectral shadow.}$$

The finger tells us where to look: not at a free-standing Gibbs measure, not at an unconstrained symbolic shift, and not at speculative global zeta functions, but at the finite residue-height spectral shadows of deterministic orbits and the missing inverse theorem that would force every bad orbit to cast one.